SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO1 ASSESSMENT TOOL 2 – Agile Sprint Planning & User Story Workshop

Course Code:	CSA10 / CSA1005	Course Title:	Software Engineering
CO Assessed:	CO1	Assessment:	Assessment Tool 2 – Agile Sprint Planning & User Story Workshop
Weightage:	20%	Total Marks:	25
CO1 Statement:	*Apply advanced methodologies and agile project management techniques to manage software projects effectively*		
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Date:	14-10-2026	Duration:	60 minutes

Code of Conduct

I *KARNAM HARSHITH - 192511384* *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: _____

Annexure A – Agile Sprint Planning & User Story Workshop

Instructions to Students:

Each student is assigned an individual software project problem statement. Based on the given problem statement, perform an Agile Sprint Planning and User Story Workshop by applying Agile software engineering practices. Prepare a comprehensive sprint planning document covering the following aspects:

Based on the given problem statement, perform an Agile Sprint Planning and User Story Workshop by applying Agile software engineering practices. Prepare a comprehensive sprint planning document covering the following aspects:

1. Identify the Project Vision and Stakeholders
   - Define the project's vision, objectives, and identify all key stakeholders involved.
2. Create Epics and User Stories
   - Break down the requirements into Epics and formulate well-structured User Stories using the format: *As a <user>, I want <feature>, so that <benefit>.*
3. Define Acceptance Criteria
   - Write clear and measurable acceptance criteria for each selected user story using the Given–When–Then format or equivalent.
4. Prioritize the Product Backlog
   - Organize and prioritize the product backlog based on business value, customer needs, and project dependencies.
5. Plan the Sprint Backlog
   - Select high-priority user stories for the first sprint and divide them into actionable development tasks.
6. Estimate Effort Using Story Points
   - Assign story points to each selected user story using an appropriate estimation technique (e.g., Fibonacci sequence or Planning Poker) and justify your estimates.
7. Present the Sprint Goal and Expected Deliverables
   - Define a clear sprint goal and describe the expected sprint deliverables, including the features to be completed at the end of the sprint.

Rubrics for Calculation

Criteria	Excellent (5 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Project Vision & Stakeholder Identification (30%)	Project vision is clear, comprehensive, and aligned with the problem statement. All relevant stakeholders and their roles are correctly identified.	Vision is clear with minor omissions. Most stakeholders are correctly identified.	Vision is partially defined. Some stakeholders are missing or incorrectly identified.	Vision is unclear or incomplete. Stakeholders are poorly identified or missing.
Epics & User Stories (25%)	Epics are well-defined and user stories are complete, follow Agile format, and fully capture functional requirements.	Epics and user stories are mostly complete with minor formatting or requirement issues.	User stories are partially complete with limited Agile structure.	User stories are incomplete, unclear, or do not follow Agile practices.
Acceptance Criteria (20%)	Acceptance criteria are specific, measurable, testable, and aligned with all user stories.	Acceptance criteria are mostly clear and testable with minor gaps.	Acceptance criteria are partially defined and lack clarity or completeness.	Acceptance criteria are missing or not aligned with user stories.
Product Backlog Prioritization & Sprint Planning (15%)	Product backlog is logically prioritized, sprint backlog is realistic, and task breakdown is complete.	Backlog prioritization and sprint planning are mostly appropriate with minor issues.	Prioritization or sprint planning is partially complete with limited justification.	Backlog prioritization and sprint planning are poorly organized or incomplete.
Story Point Estimation, Sprint Goal & Presentation (10%)	Story point estimation is well justified, sprint goal is clear, deliverables are realistic, and presentation is professional and well organized.	Estimation and sprint goal are mostly appropriate with minor inconsistencies. Presentation is clear.	Estimation or sprint goal lacks justification. Presentation is adequate but incomplete.	Estimation is unrealistic or missing. Sprint goal, deliverables, and presentation are unclear or incomplete.

Criteria	Mark
Project Vision & Stakeholder Identification (30%)	/5
Epics & User Stories (25%)	/5
Acceptance Criteria (20%)	/5
Product Backlog Prioritization & Sprint Planning (15%)	/5
Story Point Estimation, Sprint Goal & Presentation (10%)	/5
TOTAL	/25

1. Project Vision and Stakeholder Identification

1.1 Project Vision

The vision of the Intelligent IT Help Desk and Service Management Platform is to provide a centralized platform for managing IT support services within an organization. The system will allow employees to report IT problems through support tickets and enable IT teams to efficiently categorize, prioritize, assign, track, and resolve those issues.

The platform aims to replace fragmented methods such as email, phone calls, and direct communication with an organized ticket-based system. It will improve the visibility of IT requests, reduce response and resolution time, support SLA monitoring, and provide better IT service management.

1.2 Project Objectives

- To provide a centralized platform for reporting IT problems.
- To allow employees to create and track support tickets.
- To categorize problems such as hardware, software, network, and account issues.
- To prioritize tickets according to urgency and impact.
- To assign tickets to the appropriate IT support team.
- To monitor Service Level Agreements (SLA).
- To provide a knowledge base for common IT problems.
- To collect user feedback after issue resolution.
- To provide analytics and reports for IT management.
- To improve the overall efficiency and quality of IT support services.

1.3 Stakeholder Identification

Stakeholder	Role
Employees / End Users	Report IT problems, create tickets, track ticket status, and provide feedback.
IT Support Staff	Receive assigned tickets, investigate problems, update ticket status, and resolve issues.
IT Administrator	Manage users, roles, support teams, categories, system configuration, and access permissions.
IT Manager	Monitor support activities, ticket status, SLA performance, resolution time, and team performance.
Organizational Management	Review IT service reports, identify recurring problems, and make decisions for improving IT services.

2. Epics and User Stories

For the Intelligent IT Help Desk and Service Management Platform, the major requirements are divided into six epics. Each epic represents a major functional area of the system, and the requirements are expressed as Agile user stories using the required format.

Epic 1 – User Authentication and Role Management

ID	User Story
US01	As an employee, I want to securely log in to the system, so that I can access IT support services.
US02	As an IT administrator, I want to manage user roles and permissions, so that users can access only the functions appropriate to their roles.

Epic 2 – IT Ticket Management

ID	User Story
US03	As an employee, I want to create an IT support ticket, so that I can report my technical problem to the IT team.
US04	As an employee, I want to view my ticket status, so that I can track the progress of my reported issue.
US05	As an IT support staff member, I want to update ticket status and resolution details, so that the progress of the issue is properly recorded.

Epic 3 – Intelligent Categorization and Prioritization

ID	User Story
US06	As an IT support staff member, I want tickets to be categorized as hardware, software, network, or account issues, so that problems can be handled appropriately.
US07	As an IT support staff member, I want tickets to be prioritized according to urgency and impact, so that critical problems are handled first.

Epic 4 – Support Team Assignment and SLA Management

ID	User Story
US08	As an IT manager, I want tickets to be assigned to the appropriate support team, so that problems reach the correct technical staff.
US09	As an IT manager, I want to monitor SLA deadlines, so that delayed tickets can be identified and escalated.

Epic 5 – Knowledge Base and User Feedback

ID	User Story
US10	As an employee, I want to search the knowledge base for common IT problems, so that I can find solutions without always contacting IT support.
US11	As an IT support staff member, I want to add troubleshooting solutions to the knowledge base, so that commonly used solutions can be reused.
US12	As an employee, I want to provide feedback after my ticket is resolved, so that the quality of IT support can be evaluated.

Epic 6 – Analytics and Reporting

ID	User Story
US13	As an IT manager, I want to view ticket statistics, so that I can monitor the performance of IT support services.
US14	As an IT manager, I want to view SLA compliance and resolution-time reports, so that I can identify service-performance problems.
US15	As organizational management, I want to view IT service trends and recurring problems, so that I can make better decisions for improving IT services.

Epic Summary

Epic	Main Purpose
Epic 1	Secure access and role management
Epic 2	Creation, tracking and resolution of IT tickets
Epic 3	Categorization and prioritization of problems
Epic 4	Team assignment and SLA monitoring
Epic 5	Self-service knowledge and feedback
Epic 6	Performance analytics and management reporting

3. Acceptance Criteria

Acceptance criteria define the conditions that must be satisfied for a user story to be considered successfully completed. For this project, the Given–When–Then format is used as specified in the AT2 instructions. CSA10-CO1 AT2 Agile Sprint Plan…

US01 – Secure Login

Given the user has valid login credentials
When the user enters the username and password
Then the system should authenticate the user and display the appropriate dashboard.

US02 – Role Management

Given an administrator is logged into the system
When the administrator assigns a role to a user
Then the system should provide access according to the assigned permissions.

US03 – Create IT Support Ticket

Given an employee is logged into the platform
When the employee enters the problem details and submits the ticket
Then the system should create a unique ticket ID and save the ticket.

US04 – View Ticket Status

Given an employee has an existing support ticket
When the employee opens the ticket
Then the system should display the current status and ticket details.

US05 – Update Ticket

Given an IT support staff member has an assigned ticket
When the staff member updates the ticket status or resolution details
Then the system should save the changes and display the updated status.

US06 – Ticket Categorization

Given a new IT support ticket has been created
When the problem details are analyzed
Then the system should classify the ticket into an appropriate category such as Hardware, Software, Network, or Account.

US07 – Ticket Prioritization

Given a ticket has been categorized
When its urgency and impact are evaluated
Then the system should assign an appropriate priority such as Critical, High, Medium, or Low.

US08 – Support Team Assignment

Given a ticket has been categorized and prioritized
When the assignment process is performed
Then the system should assign the ticket to the appropriate IT support team.

US09 – SLA Monitoring

Given a ticket has an SLA deadline
When the deadline approaches or is exceeded
Then the system should display an alert or mark the ticket as overdue.

US10 – Knowledge Base Search

Given an employee has an IT-related problem
When the employee searches the knowledge base using relevant keywords
Then the system should display matching troubleshooting solutions when available.

US11 – Add Knowledge Article

Given an authorized IT support staff member is logged in
When the staff member adds a troubleshooting solution
Then the system should save the solution as a knowledge-base article.

US12 – User Feedback

Given a support ticket has been resolved
When the employee submits feedback
Then the system should record the feedback against the corresponding ticket.

US13 – Ticket Analytics

Given the system contains ticket records
When the IT manager opens the analytics dashboard
Then the system should display relevant ticket statistics such as open, resolved, and pending tickets.

US14 – SLA and Resolution Reports

Given ticket and SLA data is available
When the IT manager requests a performance report
Then the system should display SLA compliance and resolution-time information.

US15 – IT Service Trends

Given sufficient historical ticket data is available
When organizational management views the service reports
Then the system should display recurring IT problems and service trends.

Acceptance Criteria Summary

ID	User Story	Main Acceptance Result
US01	Secure Login	User successfully authenticated
US02	Role Management	Correct permissions applied
US03	Create Ticket	Unique ticket created
US04	Ticket Status	Current status displayed
US05	Update Ticket	Ticket changes saved
US06	Categorization	Correct category assigned
US07	Prioritization	Appropriate priority assigned
US08	Team Assignment	Correct support team assigned
US09	SLA Monitoring	Deadline/overdue alert generated
US10	Knowledge Search	Relevant solution displayed
US11	Knowledge Article	Article successfully saved
US12	Feedback	Feedback recorded
US13	Analytics	Ticket statistics displayed
US14	SLA Report	SLA/resolution report displayed
US15	Service Trends	Trends and recurring issues displayed

4. Product Backlog Prioritization

The product backlog contains all the user stories identified for the Intelligent IT Help Desk and Service Management Platform. The stories are prioritized based on business value, user needs, and dependencies, as required by the AT2 instructions. CSA10-CO1 AT2 Agile Sprint Plan…

4.1 Prioritized Product Backlog

Priority	ID	Epic	User Story	Priority Level	Story Points
1	US01	User Management	As an employee, I want to securely log in to the system, so that I can access IT support services.	Must Have	3
2	US03	Ticket Management	As an employee, I want to create an IT support ticket, so that I can report my technical problem.	Must Have	5
3	US04	Ticket Management	As an employee, I want to view my ticket status, so that I can track my reported issue.	Must Have	3
4	US06	Categorization	As an IT support staff member, I want tickets to be categorized, so that problems can be handled appropriately.	Must Have	5
5	US07	Prioritization	As an IT support staff member, I want tickets to be prioritized, so that critical problems are handled first.	Must Have	5
6	US08	Assignment	As an IT manager, I want tickets to be assigned to the appropriate support team, so that problems reach the correct staff.	Must Have	5
7	US09	SLA Management	As an IT manager, I want to monitor SLA deadlines, so that delayed tickets can be identified and escalated.	Should Have	8
8	US10	Knowledge Base	As an employee, I want to search the knowledge base, so that I can find solutions to common IT problems.	Should Have	5
9	US11	Knowledge Base	As an IT support staff member, I want to add troubleshooting solutions, so that common solutions can be reused.	Should Have	3
10	US12	Feedback	As an employee, I want to provide feedback after resolution, so that IT service quality can be evaluated.	Could Have	3
11	US13	Analytics	As an IT manager, I want to view ticket statistics, so that I can monitor IT support performance.	Could Have	8
12	US14	Analytics	As an IT manager, I want to view SLA and resolution reports, so that I can identify service-performance problems.	Could Have	5
13	US15	Analytics	As organizational management, I want to view IT service trends, so that I can make better decisions.	Could Have	5
14	US02	User Management	As an IT administrator, I want to manage user roles and permissions, so that users access appropriate functions.	Should Have	3
15	US05	Ticket Management	As an IT support staff member, I want to update ticket status and resolution details, so that the progress is properly recorded.	Must Have	3

4.2 Prioritization Logic

The backlog is arranged according to the dependency of the system:

Login → Create Ticket → Track Ticket → Categorize → Prioritize → Assign → SLA → Knowledge Base → Feedback → Analytics

The first three stories form the basic working foundation of the platform. Categorization, prioritization, and assignment then support efficient ticket processing. SLA, knowledge-base, feedback, and analytics features can be developed after the core ticket workflow is available.

4.3 Sprint 1 Selection

For the first sprint, the highest-priority foundational stories are selected:

Sprint 1 Story	Story Points
US01 – Secure Login	3
US03 – Create IT Support Ticket	5
US04 – View Ticket Status	3
Total	11

5. Sprint Backlog Planning

Based on the prioritized product backlog, Sprint 1 will focus on developing the basic working foundation of the Intelligent IT Help Desk and Service Management Platform.

The AT2 requires the high-priority user stories to be selected for the first sprint and divided into actionable development tasks. CSA10-CO1 AT2 Agile Sprint Plan…

5.1 Sprint 1 Details

Sprint Duration: 2 Weeks
Sprint: Sprint 1
Sprint Focus: User login, IT ticket creation, and ticket tracking

Selected User Stories

ID	User Story	Story Points
US01	As an employee, I want to securely log in to the system, so that I can access IT support services.	3
US03	As an employee, I want to create an IT support ticket, so that I can report my technical problem.	5
US04	As an employee, I want to view my ticket status, so that I can track my reported issue.	3
	Total	11

5.2 Task Breakdown

US01 – Secure Login

Development Tasks:

1. Design the login interface.
2. Create the user database structure.
3. Implement username and password authentication.
4. Implement basic role-based access.
5. Implement login validation and logout.
6. Test valid and invalid login scenarios.

US03 – Create IT Support Ticket

Development Tasks:

1. Design the ticket creation form.
2. Create the ticket database structure.
3. Add problem description and required details.
4. Generate a unique ticket ID.
5. Implement ticket submission.
6. Validate ticket information.
7. Test ticket creation and storage.

US04 – View Ticket Status

Development Tasks:

1. Design the ticket-status interface.
2. Retrieve ticket information from the database.
3. Display ticket ID and problem details.
4. Display the current ticket status.
5. Display basic ticket history.
6. Test ticket tracking functionality.

5.3 Sprint Backlog

ID	User Story	Main Tasks	Points	Status
US01	Secure Login	UI, authentication, roles, validation, testing	3	To Do
US03	Create Ticket	Form, database, ticket ID, submission, validation	5	To Do
US04	Track Ticket	Status page, database retrieval, history, testing	3	To Do

6. Story Point Estimation

For estimating the effort of the Sprint 1 user stories, the Fibonacci sequence is used:

1, 2, 3, 5, 8, 13

Story points represent the relative effort, complexity, and uncertainty involved in completing a user story. The AT2 instructions specifically allow Fibonacci estimation or Planning Poker. CSA10-CO1 AT2 Agile Sprint Plan…

6.1 Story Point Estimation for Sprint 1

ID	User Story	Story Points	Justification
US01	Secure Login	3	Requires login interface, authentication, validation, and basic role access.
US03	Create IT Support Ticket	5	Requires ticket form, database storage, ticket ID generation, validation, and testing.
US04	View Ticket Status	3	Requires retrieving ticket information and displaying its current status and history.
	Total	11	

6.2 Justification

US01 – 3 Points:
The login functionality is relatively simple but requires authentication, validation, and testing.

US03 – 5 Points:
Creating a ticket involves multiple components such as the user interface, database, unique ticket ID, data validation, and submission process, making it more complex.

US04 – 3 Points:
Ticket tracking mainly requires retrieving stored ticket information and displaying its status, so it has moderate complexity.

Estimation Method

The team can use Planning Poker to discuss the complexity of each story and agree on the final points. The Fibonacci values help distinguish between stories with increasing levels of effort and uncertainty.

Sprint 1 Capacity = 11 Story Points

7. Sprint Goal and Expected Deliverables

The objective of Sprint 1 is to develop the basic working foundation of the Intelligent IT Help Desk and Service Management Platform. The AT2 requires a clear Sprint Goal and expected deliverables describing the features to be completed by the end of the sprint. CSA10-CO1 AT2 Agile Sprint Plan…

7.1 Sprint Goal

"To develop a basic working IT Help Desk platform that allows employees to securely log in, create IT support tickets, and track the status of their reported issues."

7.2 Expected Deliverables

At the end of Sprint 1, the following deliverables are expected:

1. Functional Login Module – Employees can securely log in to the platform.
2. User Authentication – Valid user credentials are verified before access.
3. Ticket Creation Module – Employees can submit their IT problems through a ticket form.
4. Unique Ticket ID – Each newly created ticket receives a unique identification number.
5. Ticket Database – Submitted ticket information is stored properly.
6. Ticket Tracking Module – Employees can view the current status of their tickets.
7. Basic Ticket History – Previous updates related to a ticket can be viewed.
8. Testing – Login, ticket creation, and ticket tracking functions are tested.
9. Working Sprint Prototype – The completed Sprint 1 features work together as a basic functional system.

7.3 Sprint 1 Final Outcome

The expected workflow after completing Sprint 1 is:

Employee Login → Create IT Ticket → Receive Ticket ID → View Ticket Status

Definition of Done

A user story will be considered completed when:

- The feature is developed.
- Required data is stored correctly.
- Basic validation is implemented.
- Functional testing is completed.
- No major errors remain.
- The feature works as part of the Sprint 1 prototype.

Sprint 1 Summary

Item	Details
Sprint	Sprint 1
Duration	2 Weeks
Goal	Build the basic IT Help Desk foundation
Stories Selected	US01, US03, US04
Story Points	11
Main Deliverables	Login, Ticket Creation, Ticket Tracking
Expected Outcome	Working basic IT Help Desk prototype